Course Title	**Data Mining and Machine Learning**		**L T P C Periods**
Course Code	**UDSC-502**		**3 0 1 4 70**
Course Category	**IDM**		
Prerequisites: UDSC-101/UMAT-101, UDSC-102, UDSC-201, UDSC-203 or Equivalent			
Unit	**Title**	**Contents**	**Periods**
1	Introduction	Data Matrix, Numeric Attributes, Categorical Attributes, Graph Data, Kernel Methods, High-dimensional Data, Dimensionality Reduction, Types of Machine Learning	8
2	Regression	Linear Regression, Logistic Regression, Neural Networks, Regression Evaluation	8
3	Classification	Probabilistic Classification, Decision Tree Classifier, Linear Discriminant Analysis, Support Vector Machines, Classification Assessment.	12
4	Clustering	Representative-based Clustering, Hierarchical Clustering , Density-based Clustering, Spectral and Graph Clustering, Clustering Validation	14
5	Frequent pattern mining	Itemset Mining, Summarizing Itemsets Sequence Mining, Graph Pattern Mining.	14
		Total	56

Key Text(s):

1. Mohammed J. Zaki, Wagner Meira, Jr., Data Mining and Machine Learning: Fundamental ‘ Concepts and Algorithms, 2nd Edition, Cambridge University Press, March 2020.
2. Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman, The Elements of Statistical Learning, Second Edition, Springer, 2009
   Retrieved on July 05, 2022 from https://hastie.su.domains/Papers/ESLII.pdf

4. Chrisopher M Bishop, Pattern Recognition and Machine Learning, Springer, 2006
5. Rogers and Girolami, A First Course in Machine Learning, Chapman and Hall/CRC, 2015

References:

1. Barber, Bayesian Reasoning and Machine Learning, Cambridge University Press, 2012
2. Hal Daumé III, A Course in Machine Learning, e-Edition, 2012
3. Mitchell, Machine Learning, McGraw Hill, 1997
4. CS229-Machine Learning, Stanford University, Retrieved June 30, 2022 from https://see.stanford.edu/Course/CS229

UDSC-502: Practicals- Data Mining and Machine Learning

Unit	Title	Contents	Periods
1	Introduction	Review numpy, pandas and matplotlib for statistical, Probabilistic, algebraic and geometric views of data matrix Implementation of Linear algorithms, Non-Linear Algorithms and ensembling algorithms, bias-variance tradeoff, from scratch in python	12
2	Python Library scikit-learn	Develop different ML algorithms using scikit-learn	16

Key Text(s):

1. Jason Brownlee, Master Machine Learning Algorithms, Discover How They Work and Implement them from Scratch, Machine Learning Mastery, eBook, 2017 Retrieved June 30, 2022 https://github.com/AmandaZou/master-machine-learning-algorithms
2. Aurélien Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'reilly, 2019
3. Scikit-learn Documentation, retrieved June 30, 2022 from scikit-learn Tutorials — scikit-learn 1.0.2 documentation